

# Applied Bayesian Statistics in Animal & Biological Sciences

## 1) Introduction to Bayesian reasoning in Animal Biosciences

Henk J. van Lingen

*hvanling@uoguelph.ca*

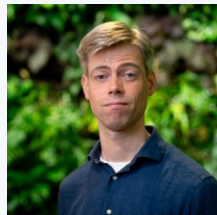
May 11, 2026



# Introduction to myself

## My positions now and previously

- PhD on (dynamic) modeling of ruminal methane production and anaerobic microbial physiology
- I also worked as a postdoc in Computational Systems Biology
- I'm currently a postdoc at the Centre for Nutrition Modelling with Jen Ellis



# Introduction to myself

## My positions now and previously

- PhD on (dynamic) modeling of ruminal methane production and anaerobic microbial physiology
- I also worked as a postdoc in Computational Systems Biology
- I'm currently a postdoc at the Centre for Nutrition Modelling with Jen Ellis



## I have expertise in (Bayesian) Statistics

- I'm an experienced biological systems modeler, I'm not a statistician
- Disclaimer: I may not be able to exhaustively explain all concepts used in this course that I took from papers or books

## Introduction of the 24 course participants

- Are you a master student, PhD candidate, or do you hold a PhD already?

## Introduction of the 24 course participants

- Are you a master student, PhD candidate, or do you hold a PhD already?
- Who has used Bayesian statistics before? WinBUGS, MCMCglmm, BGLR, brms, STAN, PROC MCMC

# Introduction of the 24 course participants

- Are you a master student, PhD candidate, or do you hold a PhD already?
- Who has used Bayesian statistics before? WinBUGS, MCMCglmm, BGLR, brms, STAN, PROC MCMC
- Who works with R, Python, Julia or SAS?

## Introduction of the 24 course participants

- Are you a master student, PhD candidate, or do you hold a PhD already?
- Who has used Bayesian statistics before? WinBUGS, MCMCglmm, BGLR, brms, STAN, PROC MCMC
- Who works with R, Python, Julia or SAS?
- What reasons did you have to sign up for this course?

## Bayesian introduction to the 24 course participants

- Who has used:
  - Mixed models

## Bayesian introduction to the 24 course participants

- Who has used:
  - Mixed models
  - Non-linear regression

## Bayesian introduction to the 24 course participants

- Who has used:
  - Mixed models
  - Non-linear regression
  - Discrete data models

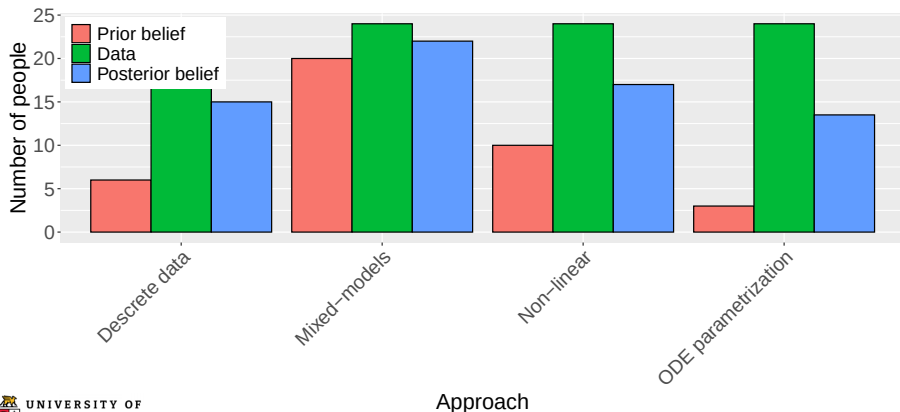
## Bayesian introduction to the 24 course participants

- Who has used:
  - Mixed models
  - Non-linear regression
  - Discrete data models
  - Parameter estimation of an ODE model?

# Bayesian introduction to the 24 course participants

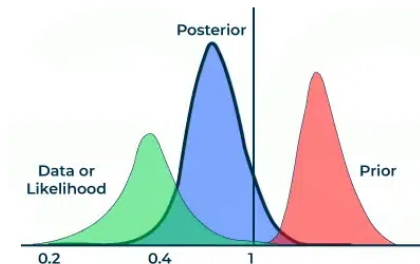
- Who has used:

- Mixed models
- Non-linear regression
- Discrete data models
- Parameter estimation of an ODE model?

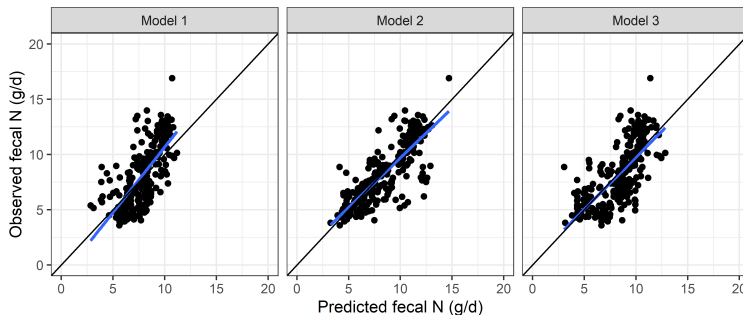


# Today's program

- Bayesian statistics examples in Animal science
- Frequentist statistics vs Bayesian statistics
- Bayesian reasoning
- The Bayes' rule



# Application in Sheep Nutrition

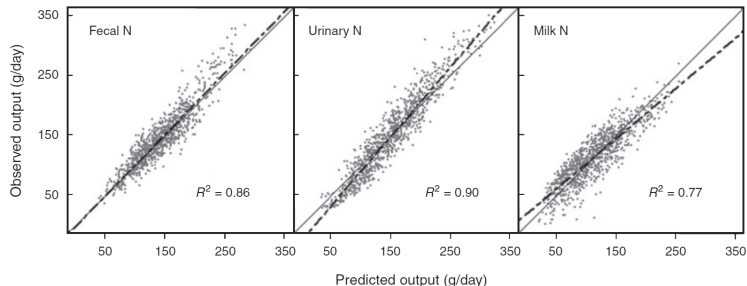


$$y_{ij} = x_{ij}^T \beta + s_i + e_{ij} \text{ with } s_i \sim N(0, \sigma_s^2), e_{ij} \sim N(0, \sigma_e^2) \quad (1)$$

| Model                                                                                                                        | RMSPE |
|------------------------------------------------------------------------------------------------------------------------------|-------|
| 1] $-1.16 \pm 0.90 + 5.72 \pm 0.64 \times \text{DMI} + 0.196 \pm 0.026 \times \text{CP}$                                     | 24.5% |
| 2] $-4.36 \pm 0.90 + 6.03 \pm 0.48 \times \text{DMI} + 0.109 \pm 0.021 \times \text{CP} + 0.115 \pm 0.008 \times \text{NDF}$ | 23.7% |
| 3] $12.3 \pm 1.0 + 5.88 \pm 0.43 \times \text{DMI} - 0.166 \pm 0.009 \times \text{DMD} + 0.123 \pm 0.018 \times \text{CP}$   | 19.3% |

Van Lingen et al., 2021

# Application in Dairy Nutrition - Predicting N excretion



$$Y_{ijk} \sim N_3(\mu_{ijk}, \Sigma) \quad (2)$$

$Y_{ijk}$  contains N excretion for  $k$  records,  $j$  animals and  $i$  studies,  $\mu_{ijk}$  a function of a matrix of covariates  $X_{ijk}$ ,  $\Sigma$  residual (co)-variance matrix

Reed et al., 2014

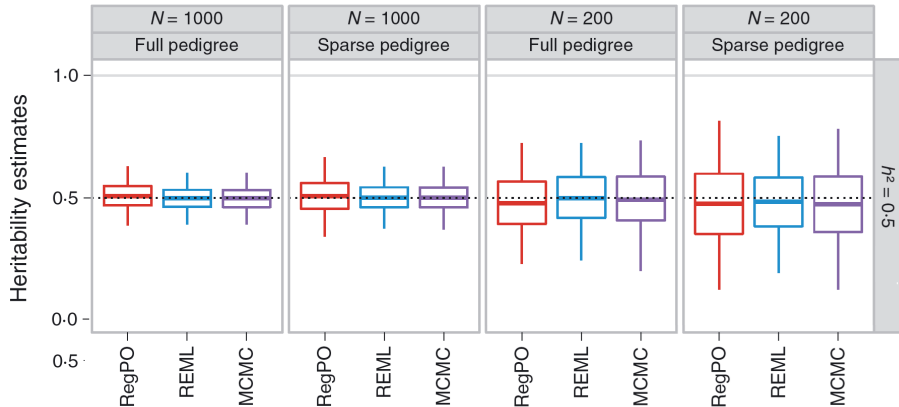
# Application in Dairy Nutrition - Multivariate N prediction

$$\Sigma_1 = \begin{bmatrix} \sigma_{\text{FN}}^2 & 0 & 0 \\ 0 & \sigma_{\text{UN}}^2 & 0 \\ 0 & 0 & \sigma_{\text{MN}}^2 \end{bmatrix}$$

$$\Sigma_2 = \begin{bmatrix} \sigma_{\text{FN}}^2 & \sigma_{\text{FN}}\sigma_{\text{UN}} & \sigma_{\text{FN}}\sigma_{\text{MN}} \\ \sigma_{\text{FN}}\sigma_{\text{UN}} & \sigma_{\text{UN}}^2 & \sigma_{\text{UN}}\sigma_{\text{MN}} \\ \sigma_{\text{FN}}\sigma_{\text{MN}} & \sigma_{\text{UN}}\sigma_{\text{MN}} & \sigma_{\text{MN}}^2 \end{bmatrix}$$

- The non-zero off-diagonal case  $\Sigma_2$  quantifies residual variance correlation, whereas the zero off-diagonal case  $\Sigma_1$  does not

# Application in Breeding & Genetics - Heritability



- Linear mixed model to assess the genetic variance component
- Similar frequentist (RegPO, REML) and Bayesian (MCMC) inference?

## Frequentist $P$ -values as a measure of evidence

$P$ -value is not the probability  $H_0$  is false ...

- Probability of observing the result or more extreme result under  $H_0$
- Then a  $P$ -value is also based on "non-observed" data

## Frequentist $P$ -values as a measure of evidence

$P$ -value is not the probability  $H_0$  is false ...

- Probability of observing the result or more extreme result under  $H_0$
- Then a  $P$ -value is also based on "non-observed" data
- Example: CH<sub>4</sub> from cattle fed control and inhibitor diets:
  - $H_0 : \Delta = \mu_{\text{CH}_4, \text{control}} - \mu_{\text{CH}_4, \text{inhibitor}} = 0$
  - $\hat{\Delta} = 60 \text{ g/d}$  may yield  $t_{\text{obs}} = 2.19$  with a  $P$ -value for the probability  $t_{\text{obs}} \geq 2.19$ , given that  $H_0$  is true

## Frequentist $P$ -values as a measure of evidence

$P$ -value is not the probability  $H_0$  is false ...

- Probability of observing the result or more extreme result under  $H_0$
- Then a  $P$ -value is also based on "non-observed" data
- Example: CH<sub>4</sub> from cattle fed control and inhibitor diets:
  - $H_0 : \Delta = \mu_{\text{CH}_4, \text{control}} - \mu_{\text{CH}_4, \text{inhibitor}} = 0$
  - $\hat{\Delta} = 60 \text{ g/d}$  may yield  $t_{\text{obs}} = 2.19$  with a  $P$ -value for the probability  $t_{\text{obs}} \geq 2.19$ , given that  $H_0$  is true

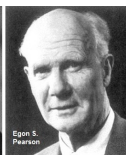
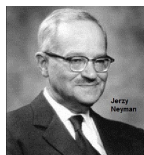
$P$ -values are not effect sizes ... :

# Frequentist paradigms: 95% confidence interval instead?

- Fisher: evidence to reject  $H_0$  according to significance level  $\alpha \leq 0.05$
- Neyman-Pearson:  $H_0$  vs  $H_1$  decision, minimize Type I (false positive) and Type II (false negative) errors, perform likelihood-ratio-test



VS

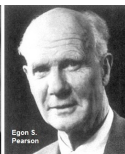
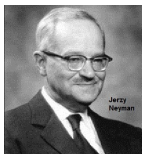


# Frequentist paradigms: 95% confidence interval instead?

- Fisher: evidence to reject  $H_0$  according to significance level  $\alpha \leq 0.05$
- Neyman-Pearson:  $H_0$  vs  $H_1$  decision, minimize Type I (false positive) and Type II (false negative) errors, perform likelihood-ratio-test



VS



Instead, consider the Bayesian view on probability:

- There is a 95% chance that the true effect size is in the *credible interval*
- One can even accompany the credible interval with the full distribution of the true effect size



## Idea of Bayesian reasoning in daily life

## Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)

## Idea of Bayesian reasoning in daily life

## Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)



# Idea of Bayesian reasoning in daily life

## Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)



# Idea of Bayesian reasoning in daily life

## Visiting Belgium for the first time:

- You may hardly know anything about Belgians (no prior knowledge)
- Travel books tell you Belgians produce excellent beers and chocolate (prior knowledge)
- You meet Belgians during your visit (data)
- You learn the Belgian cuisine provides various delicious savoury dishes and you update your opinion (posterior knowledge)



# Basic idea of the Bayesian approach: trial example

Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water



# Basic idea of the Bayesian approach: trial example

## Check the efficacy of a new mouthwash:

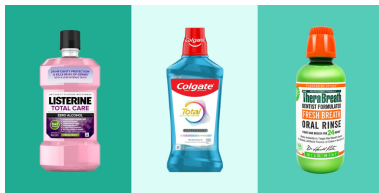
- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%



# Basic idea of the Bayesian approach: trial example

## Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%
- Use this past experience as prior knowledge and combine with the (new) data



# Basic idea of the Bayesian approach: trial example

## Check the efficacy of a new mouthwash:

- First result: 25% plaque reduction [10%, 40%] compared to tap water
- Result seems great, but previous products resulted in reductions of only 5% to 15%
- Use this past experience as prior knowledge and combine with the (new) data
- Update your knowledge on the efficacy of the new mouthwash



# The Bayesian approach - Bayes' theorem or Bayes' rule

Suppose: Event A is a positive diagnostic, B a diabetic patient

## Elementary property in probability theory:

- $P(A, B) = P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$ 
  - Joint probability of A and B:  $P(A, B)$ , marginal probabilities:  $P(A)$  and  $P(B)$ , conditional probabilities:  $P(A|B)$  and  $P(B|A)$

## Bayes' rule follows from elementary property:

- $P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$ 
  - Note:  $P(A|B)$  quantifies the probability of a positive diagnostic given that the patient is diabetic

## Verify/apply the Bayes' rule using Boston city hospital data

| Test  | Diabetic | Non-diabetic | Total |
|-------|----------|--------------|-------|
| +     | 56       | 49           | 105   |
| -     | 14       | 461          | 475   |
| Total | 70       | 510          | 580   |

$$\bullet P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

## Verify/apply the Bayes' rule using Boston city hospital data

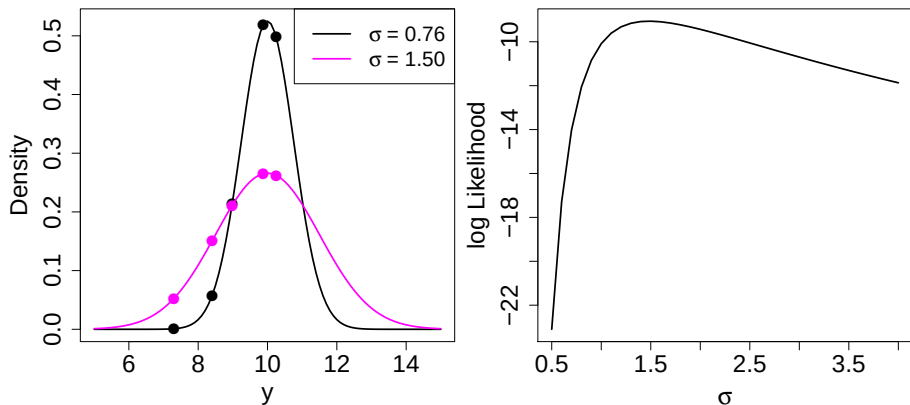
| Test  | Diabetic | Non-diabetic | Total |
|-------|----------|--------------|-------|
| +     | 56       | 49           | 105   |
| -     | 14       | 461          | 475   |
| Total | 70       | 510          | 580   |

$$\bullet P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{\frac{56}{70} \cdot \frac{70}{580}}{\frac{105}{580}} = \frac{\frac{56}{580}}{\frac{105}{580}} = \frac{56}{105}$$

Law of total probability:  $P(A) = P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)$

- Yields:  $P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A) = P(A|B) \cdot P(B) + P(A|B^C) \cdot P(B^C)}$
- If  $P(B) = 3\%$  worldwide,  $P(B|A) = \frac{\frac{56}{70} \cdot 0.03}{\frac{56}{70} \cdot 0.03 + \frac{49}{461} \cdot 0.97} = 0.20$

# Frequentist and Bayesian inference may use likelihood



- (log) Likelihood is the plausibility of model parameter values given the observed data:  $L(\theta|\mathbf{y}) = P(\mathbf{y}|\theta) = \prod_{i=1}^n p(y_i|\theta)$ 
  - $l(\mu = 10; \sigma = 1.50|\mathbf{y}) = \log(0.052 \times 0.21 \times 0.15 \times 0.27 \times 0.26) = -9.1$
  - $l(\mu = 10; \sigma = 0.76|\mathbf{y}) = -12.9$

## Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{p(\mathbf{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y}) \quad (3)$$

## Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

$$\overbrace{p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{p(\mathbf{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y}) \quad (3)$$

This simplifies to:

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{y})}_{\text{posterior}} \propto \underbrace{p(\mathbf{y})L(\boldsymbol{\theta}|\mathbf{y})}_{\text{prior times likelihood}} \quad (4)$$

# Model fitting: apply Bayes' theorem

For parameters that follow continuous distributions:

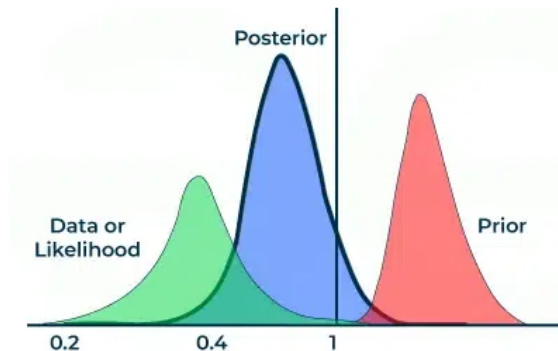
$$\overbrace{p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{p(\mathbf{y})}}^{\text{Bayes' rule}} = \frac{p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})}{\int p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta}} \propto p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y}) \quad (3)$$

This simplifies to:

$$\underbrace{p(\boldsymbol{\theta}|\mathbf{y})}_{\text{posterior}} \propto \underbrace{p(\mathbf{y})L(\boldsymbol{\theta}|\mathbf{y})}_{\text{prior times likelihood}} \quad (4)$$

- Parameters are considered random variables that may follow any distribution
- Optimizing  $p(\boldsymbol{\theta})L(\boldsymbol{\theta}|\mathbf{y})$ , i.e. data and prior knowledge, generates knowledge
- Powerful stochastic Markov-chain Monte Carlo algorithms generate the posterior

# Prior, likelihood and posterior



- Prior distribution: what you believed before seeing data
- Likelihood: what the data say is plausible/new evidence
- Posterior distribution: the updated belief after combining both

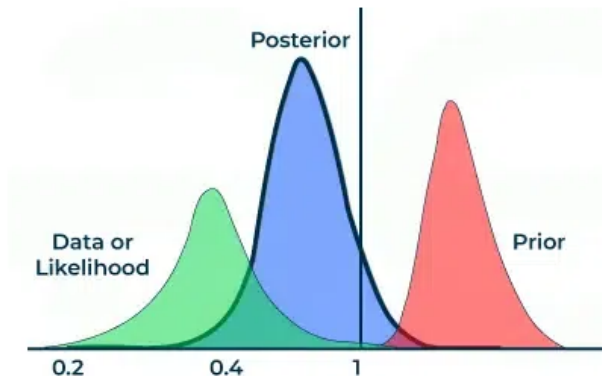
# Reasons for choosing Bayesian statistics

- ① Full Probability Statements Over Outcomes/Parameters - Not just one point value
- ② Prior usage is natural
- ③ Hierarchical (e.g. Multilevel or Mixed) Modeling Comes Naturally
- ④ Better Handling of Missing Data
- ⑤ Prediction & Decision Making
- ⑥ MCMC is a powerful optimization procedure

## Reasons for choosing Bayesian statistics

- 1 Full Probability Statements Over Outcomes/Parameters - Not just one point value
  - 2 Prior usage is natural
  - 3 Hierarchical (e.g. Multilevel or Mixed) Modeling Comes Naturally
  - 4 Better Handling of Missing Data
  - 5 Prediction & Decision Making
  - 6 MCMC is a powerful optimization procedure
- Bayesian statistics is computationally demanding
  - Bayesian statistics requires more knowledge about probability distributions

Thank you for your attention!



Questions?